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LETTER SECTION

PALEOMAGNETIC RESULTS FROM THE NAROSURA AND MAGADI VOLCANICS OF KENYA

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Results of a paleomagnetic study carried out on the exposed volcanic rocks on the western side of the South Kenya Rift Valley are presented. Nine stratigraphic groups ranging in age from Miocene to Pleistocene were sampled. The rocks consist of basalts, trachytes, nephelinites, melanephelinites, olivine melanephelinites and ignimbrites. Paleomagnetic poles obtained for different age ranges are as follows: Period I (0.64–0.72 Ma), 116°E, 85°N (A_{95} = 6°); Period II (1.6–6.9 Ma), 297°E, 84°N (A_{95} = 4°); Period III (12.0–15.0 Ma), 34°E, 80°N (A_{95} = 9°). The results for Period II show large secular variations which are in disagreement with the model predictions for near-equatorial sites.

1. Introduction

Paleomagnetic studies on the Cenozoic volcanics from East Africa have been reported by various authors. These are summarized in a paper by Reilly et al. (1976). However, there are several other areas in Kenya which have yet to be studied paleomagnetically. With a view to bridging this gap, the present authors chose the Narosura and the Magadi areas for a detailed paleomagnetic study.

2. Geology and sampling

The geological description of the areas covered by the present study is divided into two parts – the Narosura area and the Magadi area. Figure 1 shows the simplified geological structures relating to the areas. The geological sequence and the whole-rock K/Ar ages of the rock units are shown in Table I. Code letters in the second column of Table I are used in the text in order to differentiate the rock units.

2.1. Narosura area

The geology and the sampling sites of this area are shown in Fig. 2. The volcanic activities to the east of Narosura township extended from the Miocene to the end of the Pleistocene. The Kishalduga melanephelinites series (MK) rest on the sub-Miocene peneplain, and include multiple flows of melanephelinites and olivine melanephelinites (Saggerson, 1966; Wright, 1967). They erupted from fissures or vents which are now concealed. The intrusive porphyritic nephelinites (PM) outcrop in the River Sangatuny in the form of dykes. A number of flows of Lengitoto trachytes (GT) form the dissected plateaux in the northeastern corner of the area. Saggerson (1966) suggested that trachytes were fissure eruptions, probably extruded along early north–south rift faults. The olivine melanephelinites of Loisuimurto (ML) erupted from four ancient vents which are aligned along northeast–southwest directions as shown in Fig. 2. As the lava flows of these vents do not overlap, their sequence of eruption is not known.

The welded tuff (WT) forms a massive deposit in the western part of the area. It is believed to be derived from the Kishalduga volcanic activities (Saggerson, 1966), and it outcrops along the River Narosura. The ignimbrite (IG) is flinty and unbedded, and is believed to have been deposited from the Loisuimurto volcanic activities (Saggerson, 1966).

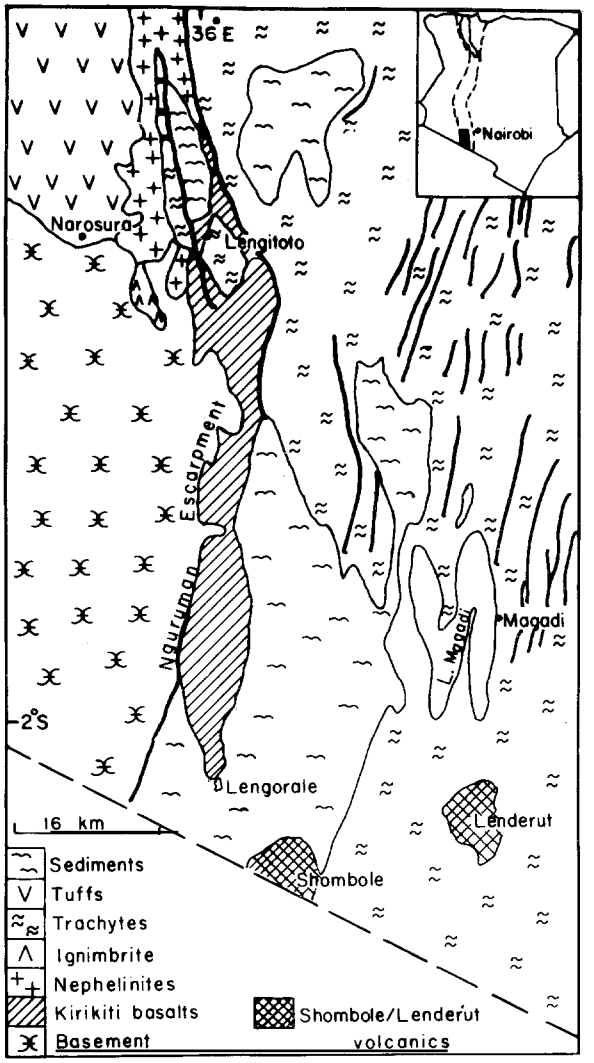


Fig. 1. A simplified map showing the geological structure of the Magadi and Narosura areas of the southwestern part of Kenya.

2.2. Magadi area

The rocks sampled from this area are Kirikiti basalts (KT), Kordjya basalts (KB), Plateau trachytes (PT) and Lengorale trachytes (LT) which outcrops mainly in the western part of Lake Magadi (Fig. 3).

The Kirikiti basalts (KT), comprising a number of flows, form a platform ~4.5-km wide in the lower part of the Nguruman Escarpment. A total of six flows of this rock unit were sampled. The lowermost flow (KT6) exposed was sampled just south of the River Oloiboroto gorge. The next three in succession (KT5–KT3) were sampled at a gorge located at 1°55′S, 36°2′E, and the two top flows (KT2 and KT1) were sampled further south of the gorge. Three flows of the Kordjya basalts (KB3–KB1) were sampled west of the Kordjya Escarpment and the other four flows (KB7–KB4) west of Little Lake Magadi. The Plateau trachytes (PT) are the most widespread and well-exposed lavas in the central area. They are believed to have welled up quietly through fissures (Baker, 1958). The trachytes are dissected by numerous sub-parallel faults, exposing one to two trachyte flows. Most of the trachytes were sampled along the road leading to the Nguruman Escarpment. The Lengorale quartz trachytes (LT), forming Ol Doinyo Lengorale Hill, rest directly on the Kirikiti basalts (KT) in the south of the Nguruman Escarpment. They are at least 30-m thick and occur as a series of massive domes.

3. Measurements

Samples were collected either as oriented blocks or as cores. Specimens of 2.2-cm diameter and 2.5-cm long were measured on a commerical version of a 5-Hz magnetometer similar to the one described by Foster (1966). Two specimens, independently oriented, from each site were progressively demagnetized in steps of 2.5 mT and the optimum demagnetizing field for cleaning was chosen using the criterion of Briden (1972). The rest of the specimens from the site were then demagnetized at this optimum field and the cleaned directions calculated.

In the analysis, the cleaned directions of several specimens were averaged to yield a single sample mean direction. The mean direction for a given site was found by averaging the directions for the several samples from that site. The site mean directions were then given unit weight in the evaluation of the mean direction of a rock unit. The mean paleomagnetic pole of the rock unit was obtained by averaging the site poles.

4. Results and discussion

4.1. Polarity, mean directions and Paleomagnetic poles

Mean directions and poles after AF demagnetization of the various rock units of the two areas are

TABLE I

Whole-rock K/Ar ages for the rock units: code letters (second column) for each unit are used in the text and figures

Rock unit	Code letters	Age (Ma)	Reference
Lengorale trachytes	LT	0.64 ± 0.09	Baker et al. (1971)
Plateau trachytes	PT	0.72–0.91	Baker and Mitchell (1976)
Kordjya basalts	KB	1.66–2.17	Baker and Mitchell (1976)
Ignimbrites	IG	2.7 ± 0.1	Crossly (1976)
Kirikiti basalts	KT	2.93–3.1	Baker and Mitchell (1976)
Olivine melanephelinites (Loisumurto area)	ML	4.2 ± 0.1	Crossly (1976)
Lengitoto trachytes	GT	6.9 ± 0.2	Crossly (1976)
Porphyritic nephelinites	PM	12.0 ± 0.3	Crossly (1976)
Welded tuff	WT	–	Crossly (1976)
Melanephelinites (Kishalduga area)	MK	15.0 ± 0.4	Crossly (1976)

shown in Table II. The Paleomagnetic directions have been classified into three categories – normal, reverse and intermediate, according to the criterion used by Dagley et al. (1967). Any direction which is not within 40 degrees absolute from the direction of the dipole field of the area has been classified as intermediate direction, and the remainder are classified as normal or reverse.

4.2. Lengitoto trachytes (GT) and olivine melanephelinites (ML)

The paleomagnetic data for the five successive flows (GT1–GT5) of the Lengitoto trachytes and for the four vents of the olivine melanephelinites of the Loisiumurto section are shown in Table II. The mean directions of the flows, with the circles of confidence,

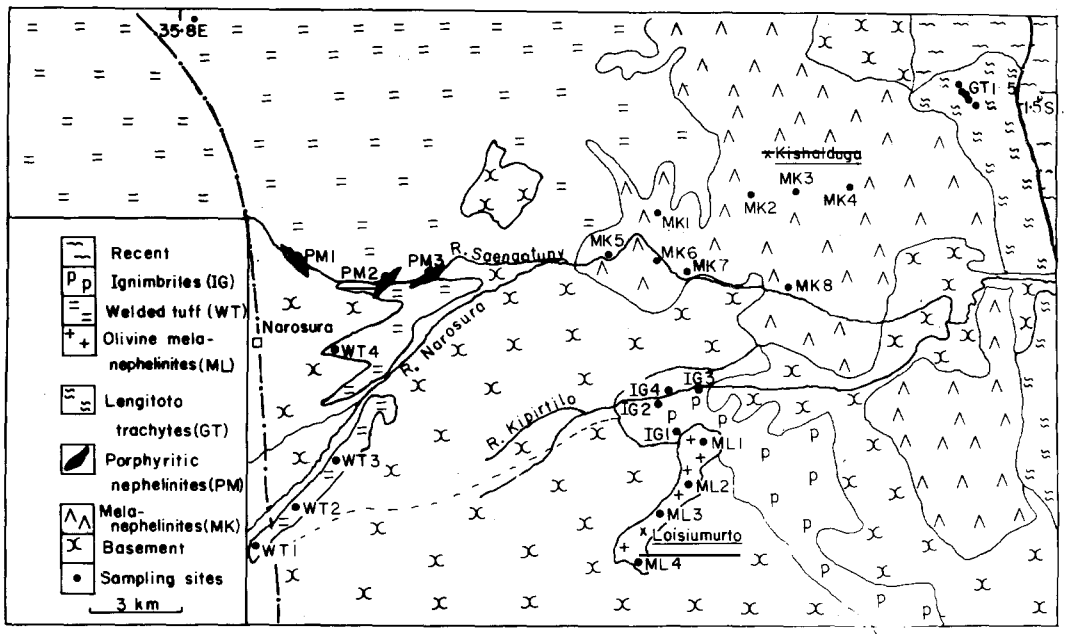


Fig. 2. Geological map of the Narosura area with the locations of sampling sites.

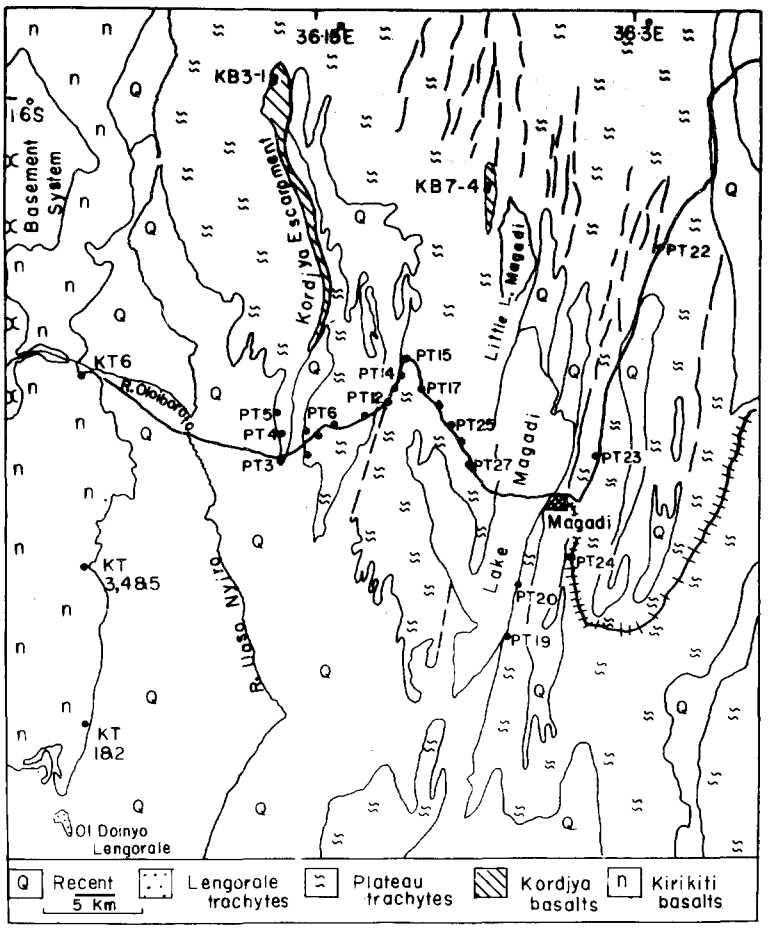


Fig. 3. Geological map of the Magadi area with the locations of sampling sites.

are plotted in Fig. 4. The NRM directions of flows GT5 and GT3 of the Lengitoto trachytes are normal and reverse, respectively, and those of flows GT1, GT2 and GT4 are intermediate (Fig. 4a). The mean direction of flow GT2 is significantly different from that of flow GT4 using the *F*-ratio test at 5% significance level. Flow GT1 has been dated at 6.9 ± 0.2 Ma. The intermediate directions are not surprising, since the time scale shows several reversals in the vicinity of 7.0 Ma (Heirtzler et al., 1968). It may be assumed that flows GT1, GT2 and GT4 consolidated when the magnetic field was in a transitional state.

No significant difference is observed between the mean directions of vents ML1 and ML3 of the olivine melanephelinites (Fig. 4b). Out of the four vents, only the lava flow of vent ML1 has been dated, giving an age of 4.2 ± 0.1 Ma, which possibly belongs to the Nunivac Event of the Gilbert Reversed Epoch (Cox, 1973). The ages of the lava flows of the other vents are not yet available. As the flows of the vents do not overlap, it is difficult to deduce the sequence of eruptions. The mean directions of the vents ML2 and ML4 are intermediate. The mean directions of vent ML4 and of flow GT1 of the Lengitoto trachytes are not significantly different.

The K/Ar ages show that the olivine melanephelinites of the Loisiumurto section and the Lengorale trachytes were consolidated at two different periods. The paleomagnetic results for these different rock units appear to be consistent with the idea that, during different polarity transitions, the field vector tends to move along the same path. There was no

TABLE II

Polarity, mean directions and paleomagnetic poles of the rock units: for abbreviations, see footnote

Rock unit	Code letters	Polarity	NS	N	HAF (mT)	After AF demagnetization					Paleomagnetic pole		
						Mean dec.	Mean inc.	RT	k	α_{95}	Long. (E)	Lat. (N)	A_{95}
Lengorale trachytes	LT	R	9	27	15.0	187.7	10.1	8.9	102.4	5.1	331.8	−81.6	4.4
		I	3	9	15.0	236.3	6.3	2.99	558.3	5.2	308.8	−31.6	4.1
Plateau trachytes	PT	R	12	19	12.5	182.7	−6.7	11.1	12.2	12.9	244.4	−84.4	10.8
		I	9	10	12.5	209.6	2.2	3.96	1.6	65.1	–	–	–
Kordjya basalts	KB	R	34(7 flows) *	60	25.0	170.1	1.5	6.6	15.4	15.9	131.4	−78.2	14.3
Ignimbrites	IG	N	3	12	30.0	345.3	−10.7	2.97	86.9	13.3	290.9	74.9	13.1
Olivine melanephelinites (Loisumurto area)	ML												
Vent ML1		R	27	31	20.0	180.4	−8.8	21.3	12.8	8.7	223.8	−84.1	8.4
Vent ML2		I	6	20	25.0	46.9	−32.1	5.8	22.5	14.4	149.0	41.9	12.2
Vent ML3		R	7	24	25.0	170.5	−21.6	5.6	12.6	19.6	177.9	−74.8	20.5
Vent ML4		I	8	30	20.0	91.2	−31.2	7.1	7.7	20.3	142.8	−0.5	23.1
Lengitoto trachytes	GT												
Flow GT1 (youngest)		I	4	18	12.5	93.3	−28.5	3.9	24.7	18.8	141.2	−3.4	19.0
Flow GT2		I	7	12	15.0	109.9	8.9	6.4	10.4	19.6	121.6	−19.9	19.1
Flow GT3		R	10	12	15.0	139.5	43.4	9.5	16.9	12.0	90.9	−44.2	12.2
Flow GT4		I	9	25	12.5	104.6	36.3	8.7	22.9	10.9	105.4	−14.2	10.9
Flow GT5		N	16	48	20.0	342.4	−38.9	15.5	30.6	6.8	254.5	63.3	7.4
Kirikiti basalts	KT	N	64(6 flows) *	53	30.0	6.3	2.8	5.3	7.3	26.6	95.1	83.7	19.1
Porphyritic nephelinites	PM	R	3	11	15.0	188.8	−12.8	2.98	171.4	9.4	263.4	−78.2	9.4
Welded tuff	WT	R	3	13	30.0	182.1	+11.1	2.97	74.3	14.4	9.9	−85.1	13.2
Melanephelinites (Kishalduga area)	MK	R	8	32	25.0	181.8	−20.9	7.52	14.7	14.9	224.2	−77.2	12.9

Abbreviations: N, number of samples; NS, number of sites; dec., declination (degrees); inc., inclination (degrees); RT, resultant vector; HAF, optimum cleaning field (mT); k, Fisher's precision parameter; A_{95}, α_{95}, radius of 95% circle of confidence (degrees); long., lat., longitude and latitude of Paleomagnetic North Pole (degrees); R, I, N, reverse, intermediate and normal, respectively.

* Unit weight given to flows.

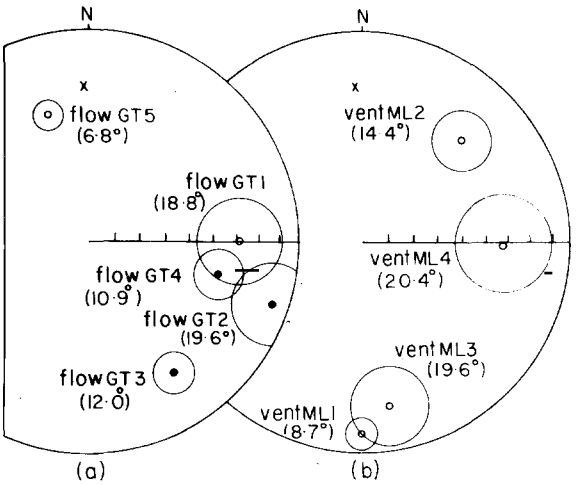


Fig. 4. Mean directions of the five flows of the Lengitoto trachytes (GT) and the four vents of the Olivine melanephelinites (ML), with circles of confidence projected on equal-area stereo nets. (•-positive dip; ○-negative dip; X-dipole field.)

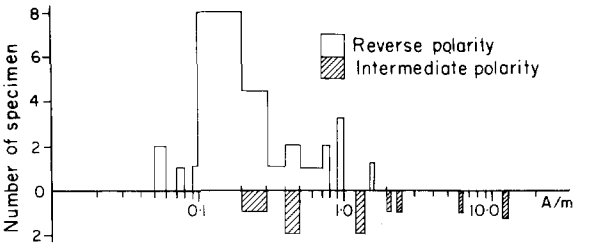


Fig. 5. Intensity histograms for the reversely and intermediately magnetized specimens of plateau trachytes.

significant variation in the intensity of the NRM among specimens of these two rock units. The average intensity for the Lengitoto trachyte specimens varied from 0.7×10^{-2} to 1.40×10^{-2} A/m, whereas for the olivine melanephelinites it varied from 1.58×10^{-2} to 5.0×10^{-2} A/m. In both the rock units, the magnetic material appears to be distributed homogeneously. No significant difference was noticed in the intensity of magnetization of samples from normally, reversely and intermediately magnetized flows.

4.3. Lengorale (LT) and plateau (PT) trachytes

Intermediate directions were observed in the Lengorale (0.64 ± 0.09 Ma) and plateau trachytes (0.72–0.91 Ma) (Table II). Declinations of the plateau trachyte specimens showed more scatter than inclinations. The intensity histogram for the reversely and intermediately magnetized specimens, Fig. 5, shows that the specimens with intermediate directions of magnetization have larger intensity spread and higher intensity values than those of the reversely magnetized specimens. The mean destructive fields (MDFs) for the reversely and intermediately magnetized specimens ranged from 10.0 to 12.5 mT.

4.4. Kordjya (KB) and Kirikiti (KT) basalts

Three flows of the Kordjya basalts from the Kordjya Escarpment and four flows from Little Lake Magadi have yielded reverse polarity. The latter flows are assumed to be older than the former, on the basis of trace elements (Crossly, 1976). The flows at the Escarpment have been dated at between 1.66 and 2.17 Ma (Baker and Mitchell, 1976). The paleomagnetic pole for the Kordjya basalts is calculated at 131°E, 78°S (A_{95} = 14°) giving unit weight to the flows. The normal polarity of the six Kirikiti basalt flows with an age range of 2.9–3.1 Ma suggests that the basalts forming the southern part of the Nguruman Escarpment erupted during the younger part of the Gauss Normal Epoch. The mean paleomagnetic pole of the six flows is calculated as 95°E, 84°N (A_{95} = 19°).

4.5. Porphyritic nephelinites (PM) and melanephelinites (MK)

The natural remanence of the porphyritic nephelinites (12.0 ± 0.3 Ma) was observed to be stable. The nephelinites are reversely magnetized. The mean direction of the three sites is Dec. = 189°, Inc. = −13° (α_{95} = 9°). In the case of the melanephelinites (15.0 ± 0.4 Ma) of the Kishalduga area, the within-site and the between-site dispersions were observed to be large compared with those of the porphyritic nephelinites (PM). The melanephelinites are reversely magnetized and the paleomagnetic pole is calculated as 224°E, 77°S (A_{95} = 13°).

4.6. Ignimbrites (IG) and welded tuff (WT)

The ignimbrites (2.7 ± 0.2 Ma) have normal polarity and belong to the Gauss Normal Epoch. The welded

TABLE III

Site mean directions, site poles and statistical parameters of rocks classified into the three periods

Period	NS	Directions					Poles			
		Mean dec.	Mean inc.	k	α_{95}	Δ *	Long. (E)	Lat. (N)	K	A_{95}
I (0.64–0.72 Ma)	21	4.9	−0.8	16.5	8.1	19.9	115.5	84.9	25.6	6.4
II (1.6–6.9 Ma)	161	354.5	−5.1	5.9	4.9	33.1	296.6	83.9	9.1	3.9
III (12.0–15.0 Ma)	14	2.2	18.8	16.7	10.0	19.8	34.3	80.1	20.8	8.9
95% Confidence limits for k and Δ										
I	$10 \leqslant k \leqslant 24, 16^\circ \leqslant \Delta < 25^\circ$									
II	$5 \leqslant k \leqslant 7, 31^\circ \leqslant \Delta < 36^\circ$									
III	$9 \leqslant k \leqslant 26, 16^\circ \leqslant \Delta < 27^\circ$									

* Δ, angular standard deviation (degrees).

tuff is undated, but is believed to be derived from the Kishalduga volcanic activities. It is reversely magnetized.

5. Pole positions and secular variation

The rock formations of the Narosura and the Magadi areas have been grouped into the following three geological periods: Period I, 0.64–0.72 Ma; Period II, 1.6–6.9 Ma; period III, 12.0–15.0 Ma. These three periods correspond roughly to Pleistocene, Pliocene and Miocene, respectively. In order to calculate the overall mean directions of the geomagnetic field for each of the periods, the reversed site directions were changed to normal polarity. The corresponding poles were then all in the Northern Hemisphere. The intermediate directions have been omitted from the analysis. The overall means of the site directions and of the site poles for each of the periods are shown in Table III.

The mean poles for Periods I, II and III are not significantly different from the present geographical pole because of the large circle of confidence observed in each of the mean poles. When compared to the mean poles, calculated from other East African volcanics of similar age groups (Reilly et al., 1976), the poles of the three periods are observed to be non-significant using the F-ratio test at 5% significance level. The Miocene pole for the lavas of the Cape Verde Islands (Watkins et al., 1968) is also in agreement with the Period III pole of the present study. The results of the present study show that the displacements of the poles from the present geographical pole increase from Period I to Period III. Due to large errors in the poles of this study, it is rather difficult to conclude whether these shifts are due to distortions of the dipole field or due to the drift of the African plate.

An attempt has been made to obtain information about paleosecular variation from the current paleomagnetic results. The simple angular standard deviation, Δ, is a widely used measured of the paleosecular variation and is given approximately by $81/\sqrt{k}$ where k is the best estimate of the precision parameter for site mean directions. The Δ values have been calculated for each of the periods, giving unit weight to sites (Table III). The confidence limits for k and Δ have been calculated using the method given by Cox (1969). The Δ values of Periods I and III are in agreement with those of the rift valley lavas in Groups C and A of 19.0 and 22.1°, respectively, (Reilly et al., 1976).

The model prediction of the equatorial value of the paleosecular variation is ~18–19° (Brock, 1971). It appears that the volcanic rocks of Period II do not record the "normal" secular variation. Since the Δ values of the other two periods agree with the equatorial value, the question arises whether the large value of Δ of Period II reflects more vigorous secular variation during Period II in Africa, or whether the secular variation has not been adequately averaged-

out. The results of Reilly et al. (1976) do not indicate whether there has been vigorous secular variation during Pliocene time. It is quite possible that the site means which were based on averages of sites with varying α_{95} may have contaminated the Δ and k values of Period II. Consequently, the possibility of vigorous secular variation may be ruled out.

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